

Topic 7 – Astronomy

Students should:	Maths skills
7.1P Explain how and why both the weight of any body and the value of g differ between the surface of the Earth and the surface of other bodies in space, including the Moon	
7.2P Recall that our Solar System consists of the Sun (our star), eight planets and their natural satellites (such as our Moon); dwarf planets; asteroids and comets	5b
7.3P Recall the names and order, in terms of distance from the Sun, of the eight planets	
7.4P Describe how ideas about the structure of the Solar System have changed over time	5b
7.5P Describe the orbits of moons, planets, comets and artificial satellites	5b
7.6P Explain for circular orbits how the force of gravity can lead to changing velocity of a planet but unchanged speed	5b
7.7P Explain how, for a stable orbit, the radius must change if orbital speed changes (qualitative only)	
7.8P Compare the Steady State and Big Bang theories	5b
7.9P Describe evidence supporting the Big Bang theory, limited to red-shift and the cosmic microwave background (CMB) radiation	
7.10P Recall that as there is more evidence supporting the Big Bang theory than the Steady State theory, it is the currently accepted model for the origin of the Universe	
7.11P Describe that if a wave source is moving relative to an observer there will be a change in the observed frequency and wavelength	5b
7.12P Describe the red-shift in light received from galaxies at different distances away from the Earth	2g 5b
7.13P Explain why the red-shift of galaxies provides evidence for the Universe expanding	5b
7.14P Explain how both the Big Bang and Steady State theories of the origin of the Universe both account for red-shift of galaxies	
7.15P Explain how the discovery of the CMB radiation led to the Big Bang theory becoming the currently accepted model	

Students should:	Maths skills
7.16P Describe the evolution of stars of similar mass to the Sun through the following stages: - a nebula - b star (main sequence) - c red giant - d white dwarf	2g
7.17P Explain how the balance between thermal expansion and gravity affects the life cycle of stars	
7.18P Describe the evolution of stars with a mass larger than the Sun	2g
7.19P Describe how methods of observing the Universe have changed over time including why some telescopes are located outside the Earth's atmosphere	